

Infusing Emotion in Text to Speech Model

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Abstract—Text-to-Speech (TTS) systems have achieved remarkable proficiency in generating natural-sounding speech. However the synthesized speech from these systems often ends up sounding robotic or machine like because it lacks the emotional qualities that humans have when they speak. Approach for emotional TTS is presented in order to overcome this constraint. Using emotion vectors taken from expressive voice samples, this method improves the text embedding by adding emotion from pre-trained models such as Tacotron. This paper adapt current TTS framework and include emotion vectors with textual input and produce emotionally complex voice output. This research project aims to make important improvements in the field of TTS synthesis, which will eventually improve the quality and naturalness of human-computer interactions.

Keywords: Text-To-Speech, Tacotron, Emotion vector, Mel-spectrogram

I. INTRODUCTION

Building a TTS system with the ability to captures emotions is a challenging task. The goal is to add speech emotion to synthesize voices, that makes synthesized voice to be more engaging and human like. Finding balance in technical accuracy and naturally expressing emotions is an ongoing effort to transform speech synthesis into more than just an informative tool to make it friendly and relatable tech for users in different situations [1, 2].

Developing a TTS technology is applicable across different domains, including robotics and automated systems, holds huge potential to give better user experience. The addition of a more human-like voice in those applications aims to produce a natural and realistic interaction, that leads to enhancing the overall work environment and user comfort[1]. Machines combined with emotional voices can make smoother user-machine interactions and have the potential to reach a broader audience, making such devices more interesting and accessible to a wide range of users.

Many computer engineers and researchers have focused their efforts on enhancing and streamlining modern TTS models, resulting in significant improvements in speech quality. Remarkable advancements have been made in the field of TTS synthesis through the application of deep neural networks (DNNs). A wide range of DNN-based

TTS models, including Char2Wav, Tacotron, Tacotron2, Fast-Speech, DeepVoice, and others, have showcased their exceptional capability to generate speech that closely mimics human-like quality. The current speech synthesis model, as shown in Figure 1, demonstrates the process of transforming input text and ground truth information into a raw spectrogram. This spectrogram is passed to post-net subsequently converted into speech using the Griffin-Lim reconstruction method.

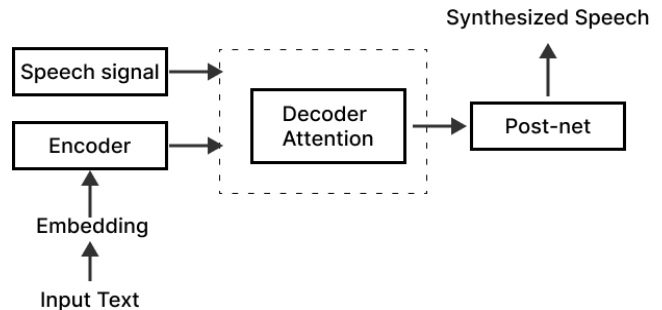


Fig. 1: Text to Speech (TTS) Model

A. Problem Defination

The infusion of emotion into TTS models poses a significant challenge in the field of artificial intelligence. While existing TTS systems excel in generating speech with clarity and coherence [5, 6], they often lack the ability to convey emotions effectively. Emotion-infused TTS aims to synthesized speech with appropriate emotional types such as happiness, sadness, excitement, or anger, contributing the naturalness and expressiveness of the generated output. This problem requires addressing various challenges, including the identification and classification of emotional content in input text, understanding the emotional expression in speech, and integrating these elements compatibly into the TTS synthesis process. Adding emotions to TTS systems needs a good understanding of language features and other speech-related concepts. It also requires developing advanced machine learning techniques that can accurately capture and reproduce emotions. This task require technical approach that combines advanced methods to model and create the small changes in speech patterns and sound

features that show specific emotional tones. Overcoming these difficulties is very important to improve the latest technology and allow for more natural and engaging interactions between humans and machines.

B. Problem Overview

The objective is to make TTS technology better by enabling it to produce speech with emotional expressions. Currently developed TTS systems cannot often convey emotions effectively, resulting in neutral and robotic speech giving low interaction with user. Infusing emotion into this system will improve the user experience and find applications in areas including in virtual assistants, audiobooks, and accessibility tools. The main idea to attach emotion vectors to text embedding before passing to next step giving text information about what type of emotion the speech contains.

II. LITERATURE SURVEY

A. Existing System

The current TTS technology is characterized by state-of-the-art models that generate speech of high quality. These models use complex neural network structures and a lots of training data to generate speech that sounds natural from the given text input. However, a key limitation of these systems is not being able to effectively convey emotions in synthesized speech [8, 9]. Despite their proficiency in replicating speech Patterns and intonations, existing models often fall in delivering emotionally expressive output, resulting in a robotic tone and limited speech style [10].

Having limitation presents a challenge in human-computer interaction, in the area of where the ability to effectively convey emotions through speech is essential like audio books. Emotional characters are important signals in communication, controlling the effectiveness of conveying meaning and engaging listeners. Without the capacity to infuse synthesized speech with emotion, existing TTS systems struggle to make long lasting connections with users, affecting their potential in various applications [11,12].

Recent advancements in TTS models [4, 6] have enabled the synthesis of a high quality speech with a neutral emotional tone, utilizing a feed forward neural network in the transformer framework to produce mel-spectrograms simultaneously, simplifying the synthesis procedure. This process involves getting alignment attention weight and context vector applying a teacher force model with an encoder-decoder structure. That predicts how long each phoneme sound should last. These attention alignments are then used by a length regulator to adjust the length of the source phoneme sequence, matching it with the target mel-spectrogram sequence. This allows the system to generate speech in parallel.

Tacotron is a sequence-to-sequence model that consists of an encoder-decoder with an attention mechanism, along

with a post-processor work together to convert text into speech. The encoder modifies input text to extract features, while the attention-based decoder focuses on different parts of the text during speech generation. The decoder uses a unique method to determine focus areas and integrates this information with previously generated content. The post-processor transforms generated features into speech waveform using the Griffin-Lim algorithm, enabling Tacotron to produce natural-sounding speech Unlike some other advanced neural TTS models like WaveNet[14], Tacotron does not rely on separately pre-trained components.

However, these models have limitations in converting textual information that combine emotional tones to speech. Addressing the expression of emotions in speech is important, particularly in applications like storybook generation and digital platform assistance as mentioned earlier. Current models struggle with precise control over the emotional content of individual words or phonemes, having a hard challenge to correctly capture word level emotional effects [15]. Leading to limitation of speech information control and may result in a mismatch with the intended context or user intentions.

In recent years there have been notable advancements in synthesizing speech that is capable to handle emotional tones. In a study conducted by researchers [18], by utilizing tacotron as base model and adding emotional information into the attention Recurrent Neural Network (RNN) by combining it with the pre-net output and adding an extra layer to adjust its size. They did the same thing in the first layer of the decoder RNN to include emotional features in the generated spectrogram. Finding out that tacotron suffers from a problem called exposure bias, which caused issues with maintaining attention alignment Monotonic Attention (MA) is implemented to solve it.

The incorporation of global style tokens (GSTs) into Tacotron by [15] improved the TTS model significantly. GSTs, trained without specific prosodic labels, showcased various expressive styles. The design of Global Style Tokens (GSTs) creates internal soft labels, giving flexibility for controlling and transferring styles. This leads to big improvements in making long speeches sound more expressive.

Research on emotion detection encompasses various neural network architectures that take into account acoustic signals and transcription. In one study, parameters extracted from audio frames contribute to the adaptation of emotional models [16]. Another investigation combines text, speech, and image inputs using Bi-LSTM, classical ML, and CNN models, with a custom-designed DCNN proving to be the most accurate [17]. A separate study focuses on prosody detection and aligning human-rated emotional content in MOOC TTS [22]. Furthermore, a Mandarin emotion speech dataset called EMOVIE is introduced alongside a model based on FastSpeech2 [19]. Lastly, a multimodal approach

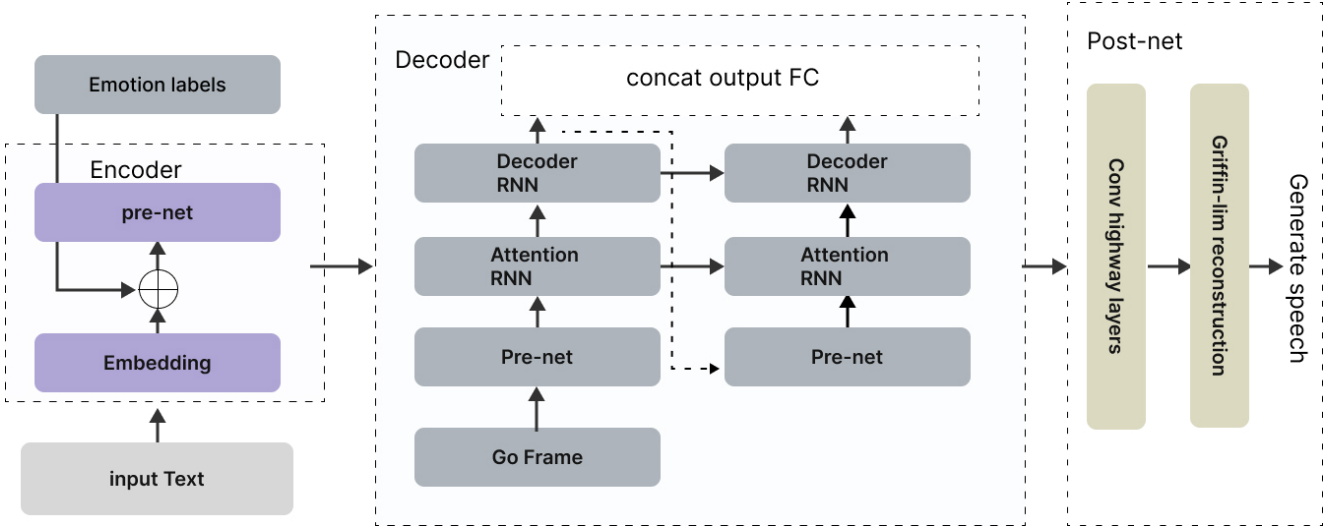


Fig. 2: Model Architecture

to speech emotion recognition is explored, where dual recurrent encoders with attention surpass other methods in capturing emotional information from audio and text modalities [20, 21].

B. Proposed System

The proposed emotional TTS model is designed to effectively convey emotions in synthesized Speech by integrating emotion vectors extracted from expressive speech samples into the text embedding process. This integration is implemented within pre-trained model Tacotron, utilizing end-to-end neural architectures. As shown in Figure 2, text and the intended emotion are supplied into the TTS Model, which produces a matching raw spectrogram with emotional expressiveness. The produced spectrogram is processed using the Griffin-Lim reconstruction, resulting in emotional speech synthesis.

The model enhances existing TTS framework by accepting emotion vectors alongside textual input during synthesis, enabling the infusion of emotional expressions into the synthesized speech output. Directly incorporating emotion vectors into the text embedding, the model can capture and convey a diverse range of emotions, ultimately improving the naturalness and expressiveness of speech.

To successfully capture and express emotions, the model's design specification to add emotion vectors into the text embedding process. Making the neural architecture to process text and emotional inputs together and pass to spectrogram generator this is achieved using tacotron as main synthesizing model that generate high quality speech. By using Mean Opinion Score (MOS) assessments to measure output that offer an organized mechanism for evaluating the expressiveness and perceived quality of the synthesized speech, enabling an unbiased comparison with current models.

III. METHODOLOGY

A. Data Collection

The Emotional Speech Database (ESD) [21] a resource which contains 350 sets of spoken sentences from 10 English and 10 Chinese speakers. These recordings capture a different type of emotions including neutral, happy, angry, sad, and surprise each having around 1 to 3 sec speech. That spans over 29 hours of content totally. The ESD stands as one of the largest and most variation multimodal speech databases currently accessible for emotional speech research.

For this study, a native English speaker is selected focusing on 5 emotional categories provided in the ESD database happy, angry, sad, surprise and neutral as experimental data.

B. Data Preprocessing

The preprocessing of the ESD dataset involves making a relation between raw audio and correspondence text then the result is stored in map like structure, after storing data text is preprocessed by removing special characters and applying tokenization this process change texts to vectors making it ready to be used for training. Text extracted from ESD contains emotional labels which is then extract to different variable to separately process emotional tokens. The audio data is then converted into linear spectrograms using librosa library this library contains useful methods for processing signal data. In the end 80-channel mel-Spectrum was extracted with a frame shift of 12.5ms and a frame length of 50ms.

To keep audio data consistent and comparable, mel-spectrum is standardized by making sure it have a zero-mean and a unit variance, serving as the standard point for further processing steps. Since the input data need to have uniform sequence length padding zeros were applied to processed data based on the maximum sequences from each batch for model training.

C. Model Architecture

The model architecture is founded on the tacotron seq2seq model, which specifically build to predict mel-spectrograms from textual inputs using teacher forcing methods and Bahdanau attention mechanism. It comprises a text encoder, an emotion encoder, and an audio decoder. The text encoder retrieves text embedding from the input text and the emotion encoder retrieves emotional embedding from emotion tokens when training and when testing emotional representations is given as inputs.

Throughout the decoding train process, the mel-spectrogram generated in the previous time step is not utilized as input for the audio decoder instead for faster training target mel-spectrogram is used as decoder input. The text embedding from the text encoder undergo an attention mechanism with the hidden state of these previous Mel Spectrograms input to the decoder. The attention values computed at this stage are used to modify the text embedding vectors and audio decoder weights.

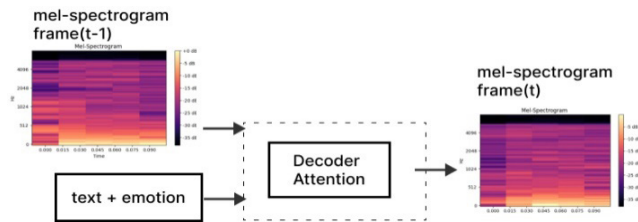


Fig. 3: Mel-spectrogram Combined Embedding

Text embedding are combined with the emotional embedding to produce text-emotion vector this is achieved by concatenating the vectors. The combined text-emotion embedding capture linguistic content from text and emotional tone specified by the emotion inputs. The audio decoder takes these combined embedding as input and uses an autoregressive process to predict the Mel Spectrograms for the current time step. The autoregressive process in figure 3 involves utilizing the combined Embedding along with the previous Mel Spectrograms input to iteratively generate the output Mel Spectrograms frame by frame.

The decoder predicted mel-spectrograms cannot be directly converted to audio to produce audio the Griffin-Lim technique is used to transfer mel-spectrogram to the time domain. By reconstructing the mel-spectrogram algorithm to a linear-spectrogram. This conversion enable model to generate an audio waveform output from the predicted mel-spectrogram, which includes the input text content and the emotion tone acquired by the combined text-emotion embedding.

It's important to note that this approach is effective during training, where the model learns to generate Mel Spectrograms conditioned on both text and emotion. However, for testing purposes, additional steps or modifications may be needed to incorporate emotional tokens and ensure accurate synthesis of emotionally rich speech.

D. Model Training

The model is trained on ESD dataset having a batch size of 32, this is done to reduce computational complexity. The dataset is sorted according to text sequence length and padded to equally sized sequences, each input is padded according to maximum sequence length in the batch. The processed texts are passed to encoder to combine linguistic features and emotional tokens. These features are sent to decoder to map textual information into frequency-time domain or spectrogram. Decoder uses attention mechanisms is to align the generated spectrograms with the input text and emotional context. Decoder weight is updated after calculating loss at each step of generated mel-spectrogram.

The training iterations were conducted over 70k steps, allowing the model to learn how to generate speech outputs conditioned on both textual inputs and emotional information. To construct the final speech output, we utilized the Griffin-Lim algorithm, refining the waveform reconstruction through 50 iterations to enhance the naturalness of the synthesized speech.

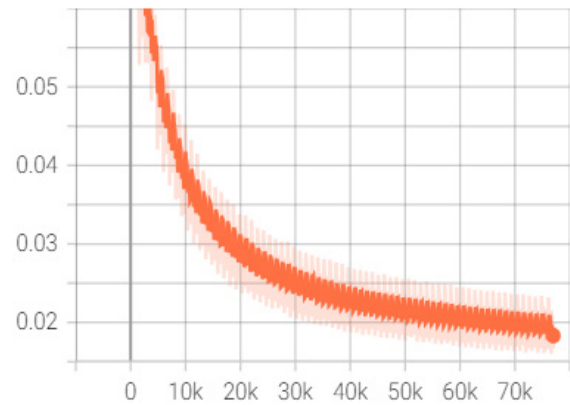


Fig. 4: Linear Spectrogram Loss

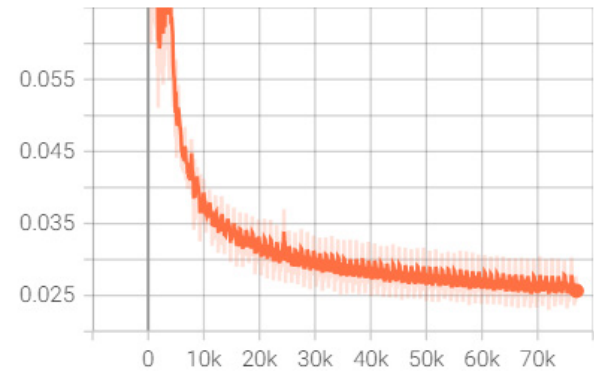


Fig. 5: Mel Spectrogram Loss

Figure 4 illustrates the training loss trend with linear spectrogram input, showing a gradual decrease in loss values as the model learns over time and Figure 5 displays the training loss pattern with mel-spectrogram input, indicating a gradual and slow convergence in loss values as training progresses.

IV. RESULT

In the listening tests, we assessed the quality of the synthesized speech using the Mean Opinion Score (MOS) as shown in Table 3. Listeners rated the quality on a scale from 1 to 5, where 5 meant excellent, 4 meant good, 3 meant fair, 2 meant poor, and 1 meant bad. We kept the linguistic content consistent across various models to prevent interference from other variables. Fifteen participants took part in these tests, each listening to five synthesized speech samples.

Table 1: Evaluation using a 5-point scale Mean Opinion Score (MOS)

Model	MOS
Emotional TTS	3.6

V. CONCLUSION

In conclusion, the modified Tacotron that has been suggested functions as a complete emotional speech synthesizer, able to produce wave patterns that correlate to specified emotions and letter sequences. Experiments revealed a considerable relationship between the sharpness and purity of the attention alignment and the generated speech quality.

We compared the outcomes with the current model in order to corroborate these findings. Although the updated Tacotron did performs same as current model, it did introduce certain fundamental alterations that might lead to future improvements. Nevertheless, we had difficulties with the synthesis process, primarily related to the Griffin-Lim reconstruction's slowness. The ultimate objective remains, despite this limitation: developing a dynamic Text-To-Speech (TTS) system capable of generating speech with multiple personalities, enhancing the ultimate objective remains, despite this limitation, developing a dynamic text-to-speech (TTS) system capable of generating speech with emotion enrich audios, enhancing user experience.

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